

DMP Evaluation Criteria

[A.sid.1] Indicates author name(s)

Scored as either 1 (fulfilled) or 0 (not fulfilled).

The name of the applicant, the author(s) of the DMP, is/are clearly indicated. This need not coincide with the person responsible for the DMP (stated later, see G.resp.5 and G.resp.6). If DMP responsibility is given to an individual, this point is still not fulfilled if there is no explicit statement that that person has also written the DMP.

Rationale: Without author names, it is impossible to know who created the DMP, or contact them.

Fulfilled Examples	Not fulfilled examples
• Author: James Red • Authors: James Red, Elizabeth Green • This document was redacted by James Red and Elizabeth Green, with contributions from all partners.	• No indication of author names. • The responsibility for the implementation and update of the DMP is given to James Red (without any other indication)

[A.sid.2] Indicates Author(s) contact information

Scored as either 1 (fulfilled) or 0 (not fulfilled).

At least one author has their full contact information clearly stated. This could also be the contact information of their institutions. Defining an ORCID record for the author automatically fulfills this criterion, as the ORCID record is usually kept updated with the current email addresses and other contact information of the author. Similar to A.sid.1, if a person is designated to be responsible for the DMP, the point is not fulfilled if they are not also explicitly stated that they have authored the DMP.

Rationale: Without author contact information, it is impossible to contact them in case comments or concerns regarding the DMP arise.

Fulfilled Examples	Not fulfilled examples
• Author: James Red (james.red@institution.org) • Authors: James Red (james.red@institution.org), Elizabeth Green • Author: James Red (contact@institution.org) • This document was redacted by James Red and Elizabeth Green, with contributions from all partners. Contact james.red@institution.org for information. • Author: James Red (ORCID: 0000-0000-0000-0001)	• Author: James Red • Authors: James Red, Elizabeth Green • The responsibility for the implementation and update of the DMP is given to James Red (james.red@institution.org) (without any other indication)

[A.sid.3] Indicates author(s) OrcID or other PIDs

Scored as either 1 (fulfilled) or 0 (not fulfilled).

At least one author as their ORCID or similar persistent identifiers clearly stated. If it is not an ORCID, the kind of PID and information on how to resolve it is also included. Acceptable kinds of PID services are up to the reviewer's judgement. Score 0 if the PID service is obscure or has the

possibility to not be permanent or promptly updated (e.g. W3C PURLs). In Similar to A.sid.1, if a person is designated to be responsible for the DMP, the point is not fulfilled if they are not also explicitly stated that they have authored the DMP.

Rationale: OrcIDs and other PIDs pinpoint with exact precision the author(s) of the document.

Fulfilled Examples	Not fulfilled examples
• Author: James Red (ORCID: 0000-0000-0000-0001) • Authors: James Red (ORCID: 0000-0000-0000-0001), Elizabeth Green • Authors: James Red (https://trustedpid.service/1234)	• Author: James Red • Author: James Red (james.red@institution.org) • Author: James Red (https://mywebsite.com/) • The responsibility for the implementation and update of the DMP is given to James Red (james.red@institution.org, ORCID: 0000-0000-0000-0001) (without any other indication)

[A.prog.1] IA - Indicates the funding program and funding body

Applies only when the DMP is written for a funded project with a specific program (e.g. Horizon Europe). Scored as 1 (fulfilled), 0 (not fulfilled) or NA (not applicable).

Clearly states the funding program type. Clearly states the organization(s) that are funding the project. Contact information for the funding agencies is not required if they are internationally recognizable (e.g. Horizon Europe from the European Research Council). Logos and other non-explicit indications could satisfy this criterium if they are sufficiently obvious and include the funding body name.

Rationale: Funding body may have an effect on data handling and sharing.

Fulfilled Examples	Not fulfilled examples
• This project was funded by the Horizon Europe program by the European Union as part of Pillar I – Excellent Science. • This project was funded by the Academy of Science (science.academy@institution.org). • Grant information can be found at https://static_grant_url.com/	• No information on funding program or funding body. • This project was funded by the Academy of Science (too generic, without contact information).

[A.ver.1] Indicates the version of DMP

Scored as 1 (fulfilled) or 0 (not fulfilled).

Clearly states the current version of the DMP. Any sensible versioning schema is accepted (e.g. 1, 2, 3, semantic versioning, etc...).

Rationale: DMPs may change over time. It is important to keep track of the current version.

Fulfilled Examples	Not fulfilled examples
• DMP version: 1 • DMP version 3.2.6	• No indication of DMP version.

[A.ver.2] Indicates the changelog of the DMP

Scored as 1 (fulfilled), 0.5 (partially fulfilled) or 0 (not fulfilled).

Includes a complete changelog of the DMP, stating each version and the changes that have occurred in the document. Can potentially be scored partially (0.5) if some versions are skipped or the detail level of the changelog is not satisfactory.

Rationale: DMPs may change over time. It is important to keep track of the changes appointed to them.

Fulfilled Examples	Partially Fulfilled Examples	Not fulfilled examples
Changelog: v1 – initial release; v2 – added information on new data kind “spectroscopy fields”; v3 – updated author affiliations	- Changelog: v1 – initial release; v3 – updated author affiliations (missing versions) - Changelog: v1 – initial release; v2 – updates; v3 – updated data section (unsatisfactory level of detail)	No DMP changelog.

[A.pid.1] Indicates a project identifier or PID

Scored as 1 (fulfilled) or 0 (not fulfilled).

Includes the project identifier (in the funding program context) or another PID related to the project. This can also be arbitrary and local (e.g. at institution level). This PID must be sufficient to keep track of the DMP throughout its various versions.

Rationale: PIDs are essential in keeping track of various aspects of the project, and it is important that the DMP contains this information for cross-referencing purposes.

Fulfilled Examples	Not fulfilled examples
Project ID: 00001234	- DMP deposited with DOI: https://doi.org/1234 (not a project-level DOI). No indication of DMP or project identifier.

[B.dtyp.1] Lists all types of gathered data

Scored as 1 (fulfilled), 0.5 (partially fulfilled) or 0 (not fulfilled).

Fulfilled if all data kinds used by the project are listed. A data “kind” is the nature of the data being collected, the phenomenon or trace that is trying to be captured. It is hard to determine if all kinds of data are actually listed when the DMP is first created. If possible, check the actual results or with the authors to see if the list is accurate or not.

Rationale: The DMP is used to plan the management of each different data type. A list of these types is essential in order to do so.

Fulfilled Examples	Partially Fulfilled Examples	Not fulfilled examples
The project will gather three types of data: aerodynamic traces of different plane models, wing shapes of different plane models, and types of engines of different plane models.	- The project is expected to gather data on the behavior of planes in different weather conditions based on their model (not enough information on each data kind) - The project will gather three types of data, one being wing shapes of various plane models (some expected data kinds are omitted).	No list of expected data types. - Collaborating institutions will gather different kinds of data relevant to the project (no list of actual data kinds)

[B.dtyp.2] Lists all formats of gathered data

Scored as 1 (fulfilled), 0.5 (partially fulfilled) or 0 (not fulfilled).

Similar to B.dtyp.1, but regarding digital data formats (e.g. .pdf, .png) or, alternatively, on the types of physical objects stored, catalogued based on type and other storage features or requirements. One preferred data format should be specified. Other formats for the same type of data can be indicated, if properly explained.

Rationale: The data formats are essential in determining how well they might be handled internally or shared externally.

Fulfilled Examples	Partially Fulfilled Examples	Not fulfilled examples
- The project will use three main data formats: .csv for tables, .wav for audio recordings and .png for images. - For data tables, they will be stored as .csv. When the format of the data is too large to be stored as .csv, .parquet data files will be used.	- Image data will be stored as .tiff, .png or .jpg (too many data formats for a single type without explanation). - Data will be stored as .csv, .wav and .png (too generic).	No list of expected data formats. - Data will be stored in the most efficient format available (too generic).

[B.dtyp.3] Lists all combinations of data type and format separately

Scored as 1 (fulfilled), 0.5 (partially fulfilled) or 0 (not fulfilled).

This point specifically refers to listing all data kinds and data format combinations, along with their descriptions. This is usually done as a table, but the format is not important.

Rationale: Each data type may be collected with different data formats. Explicitly handling different data format-and-type combination separately is important to determine the impact of these decisions precisely.

Fulfilled Examples	Partially Fulfilled Examples	Not fulfilled examples
The project will handle the following types of data: brain scans of patients during rest or exertion in voxel format (.vxl), brain scans of rats during rest or exertion in voxel format (.vxl), metadata regarding samples in tabular formats (.xlsx, .xls) and a data vocabulary as a BLOB dictionary (.dblob).	The project will handle brain scans of rats and humans in rest or exertions conditions in voxel format (.vxl) (not enough separation between types)	No list of expected data formats, no list of expected data types, or them listed separately.

[B.dtyp.4] Uses only open or widely used community formats

Scored as 1 (fulfilled) or 0 (not fulfilled).

Only open formats or widely used, common community formats are specified. A list of suggested formats for various common data types can be seen in Appendix 1 of this document. More specific formats for specialized topics must be evaluated singularly, such as by using FAIRSharing.org (<https://fairsharing.org/>) or similar services. The score given to this criterium cannot be higher than that given for B.dtyp.2. A novel data format can be considered “open” if the project also plans on sharing it as a result, but the creation of new formats must be appropriately explained as impossible to avoid. This criterium is fulfilled also if closed data types are used **during** the project, but are explicitly changed for preservation and sharing purposes (e.g. usage of Word documents that are then exported into PDF/A-1a for preservation). A short explanation of the procedure may also be included. This criterium also applies to data packaging formats, like compression algorithms.

Rationale: Open and community formats are a necessary prerogative for effective long-term preservation and interoperability.

Fulfilled Examples	Partially Fulfilled Examples	Not fulfilled examples
- The project will handle the following types of data: brain scans of patients during rest or exertion in voxel format (.vxl), brain scans of rats during rest or exertion in voxel format (.vxl), metadata regarding samples in tabular formats (.csv) and a data vocabulary as a JSON dictionary (.json). - The project will handle written documents. They will be created as Word (.docx) documents during the project but will be shared with the Open Document Format (.odf). Conversion issues will be handled appropriately.	The project will handle the following types of data: brain scans of patients during rest or exertion in voxel format (.vxl), brain scans of rats during rest or exertion in voxel format (.vxl), metadata regarding samples in tabular formats (.xlsx, .xls) and a data vocabulary as a BLOB dictionary (.dblob). (usage of some obscure or proprietary data types, .xlsx, .dblob)	No list of expected data formats, no list of expected data types, or them listed separately.

[B.dtyp.5] Explicitly justifies the use of each data format

Scored as 1 (fulfilled), 0.5 (partially fulfilled) or 0 (not fulfilled).

Each data format specified is explicitly defended by a (short) explanation, with clear references to open specifications or community guidelines. These can be omitted for overwhelmingly popular formats, like csv.

Rationale: The choice of data format must be pondered sufficiently before that particular data format can be used.

Fulfilled Examples	Partially Fulfilled Examples	Not fulfilled examples
- Brain scans will use the voxel format (.vxl). The specification of the format is available online (link) and was chosen as it is the most widely used format for these types of data. Several analysis and data processing pipelines are already compatible with this format and it is the most efficient data type available for storage purposes. - PDF files will be used to store publications and other documentation related to the project.	Descriptions are given for some, but not all data types.	No listed data formats No explanations of the choice of data formats.

[B.dtyp.6] IA - Indicates encodings and subtypes of data formats

Scored as 1 (fulfilled), 0.5 (partially fulfilled) or 0 (not fulfilled).

Lists for each data type its specific encoding (e.g. UTF-8) and (eventual) subtypes, like PDF/A-1a. The criterium is fulfilled even if these information are given only for preservation-specific data formats.

Rationale: Not all encodings or subtypes are suitable for sharing.

Fulfilled Examples	Partially Fulfilled Examples	Not fulfilled examples
PDF files will be used to store publications and other documentation related to the project. These documents will be converted to PDF/A-1a files for sharing and preservation purposes. CSV files will always be encoded as UTF-8.	Encodings and subtypes are given for some, but not all data types.	No listed data formats

[B.reuse.1] Addresses how data will be searched for reuse

Scored as 1 (fulfilled), 0.5 (partially fulfilled) or 0 (not fulfilled).

Fulfilled if at least one of the following is true: In all cases, it is up to the reviewer to decide if the description is sufficiently exhaustive or not. Full scores are given to true efforts to avoid the generation of data, or sufficiently clear explanations of why generating data is not avoidable. Partial points are given if some types of data are covered, while some other points are not.

Rationale: Re-using data is fundamental to avoid wasteful data re-generation. Additionally, a proper search strategy (i.e. setting the **desiderata** for the nature of data) informs data collection efforts and metadata choices.

Fulfilled Examples	Partially Fulfilled Examples	Not fulfilled examples
- Five major databases for transcriptomic data will be searched for RNA-seq data from HMEC-1 cell lines. Only data with at least 1M reads per sample will be suitable... (and more details, especially regarding sampling) - Five major databases have been previously searched for reusable data. In particular ... (details of data selection and quality criteria). Of the 8 datasets that matched our search parameters, none fulfilled our data quality criteria. - This project aims to expand our current body of knowledge regarding data on the tensile strengths of various materials. Generating new data is one of the main objectives of the project. - This is a pioneering project aiming to sample ice crystal formations on the north pole of Mars. No similar projects have ever been performed.	- We searched five major databases for RNA-seq data regarding HMEC-1 cell lines and found no suitable data for reuse (too little detail). - We will use 2 types of data, Y and Y'. We searched X database for Y data with Z selection criteria (only one data type was covered).	- We will search thematic databases for reusable data (not enough information on selection criteria). - We expect no data to be reusable for this project (no effort is documented to find reusable data). - No information is given on searching for existing data to reuse.

[B.reuse.2] Explains reusable data selection criteria

Scored as 1 (fulfilled), 0.5 (partially fulfilled) or 0 (not fulfilled).

Clearly explains the criteria in place that data must have to be reused. Alternatively, clearly details the discard criteria that would make data unsuitable for reuse.

Rationale: Properly determined discard criteria (i.e. setting the **desiderata** for the nature of data) informs data collection efforts and metadata choices.

Fulfilled Examples	Partially Fulfilled Examples	Not fulfilled examples
Data regarding HMEC-1 cell lines from a number of selected providers will be considered for reuse. The maximum passage number allowed will be 5. FASTQ files with at least 1M reads will be considered.	Details are not given for all data types. - Data regarding HMEC-1 cells lines will be selected if it fulfills certain quality parameters (Too generic).	No reference to specific quality criteria for selection.

[B.reuse.3] Explains why new data needs to be gathered

Scored as either 0 (not fulfilled) or 1 (fulfilled).

Clearly explains why new data must be gathered for the project. Provides evidence that previous searches for reusable data have failed and/or that specific requirements of the project require novel data to be recorded. Argues for the creation of each data type explicitly, or, alternatively, addresses novel data collection as a whole and (briefly) addresses its use during the project. Project always generate new data (even just analysis code is new data), so at least one novel data type will always be there.

Rationale: New data should be minimized. Storage, FAIRification and handling of data comes at a cost, and this cost should be reduced. This point allows the author to consciously decide if all data types are required for the project's goals.

Fulfilled Examples	Not fulfilled examples
- This project aims to monitor Macaque mating calls in modern day Gibraltar. New data must be collected. - No suitable data regarding the genetic makeup of <i>B. leonensis</i> is available, making new data collection impossible to avoid. - While bibliographies of the works of Alighieri are common, significant doubts have arisen in recent years that a significant portion of his works are omitted (including citation), making the curation of a new bibliography required. - This project uses previously published data and performs a metanalysis on it. This process is inherently novel and so is the result.	No explanation on the requirement for new data to be collected. One or more data types are not sufficiently explained. - This project requires the collection of novel data to fulfill its aims (too generic).

[B.prov.1] Explains how data provenance will be tracked

Scored as either 0 (not fulfilled), 0.5 (partially fulfilled) or 1 (fulfilled).

Clearly considers how data provenance (i.e. person that collected the data, location, time, date, processing steps, measurement device, etc...) will be recorded, often at the moment of recording. What specific information is "provenance" changes between fields, so it's up to the reviewer to decide if enough provenance is planned to allow effective reuse.

Rationale: When new data is created, its provenance is one of the most important metadata features.

Fulfilled Examples	Partially Fulfilled Examples	Not fulfilled examples
• The location, time and the characteristics of the camera apparatus used to generate the pictures are automatically saved. Authorship information will be recorded separately when images are stored in the central server. • Specific data provenance collection forms will be developed for MRI scans, will be filled as MRI scans are performed and will be saved together with the resulting data. • Code development will happen on GitHub, which automatically tracks the time and authorship of each commit.	• Details are not given for all data types. • Provenance will be tracked using specific methodology (too little information).	• No reference to data provenance.

[B.compat.1] Indicates if required software will be preserved

Scored as 1 (fulfilled) or 0 (not fulfilled).

Explicitly indicates if particular software is required to read, understand or reuse the data. If so, clearly indicates the version(s) required, the location of the source code and other information to identify the software. If containerization or other software packaging solutions are used, this information may be omitted (see B.compat.2). Common data formats expected to be readable (or easily converted) into the far future are exempt from this requirement, although a simple statement considering this problem (and dismissing it) is still required. If proprietary software which cannot be shared must be used, a strong, detailed argument for why its use is unavoidable must be provided. If multiple data types are involved, even one without its required software makes this criterium not fulfilled.

Rationale: Some specialized data types require uncommon software to be shared alongside it, or the data is inaccessible.

Fulfilled Examples	Not fulfilled examples
• This project does not use any data formats that require uncommon software to be read or understood. • The software “readVoxels” is required to visualize and process the MRI voxel data. • Usage of the proprietary format “.tub” for tracking the shape of tubercles is required as there are no other open-source alternatives available. Conversions of the format into others will be taken in consideration during the project.	• No explanation on the need to store software is given. • One or more required software are not detailed. • Software to access the data will be provided upon request (too generic).

[B.compat.2] IP - Indicates how required software will be preserved

Applicable only if specific software is required to access the data for preservation purposes (B.compat.1). If it is unknown whether software is required or not, score this as 0. Scored as 1 (fulfilled) or 0 (not fulfilled).

If specific software is required to be preserved alongside the data, the DMP specifies how this will be achieved. Judgement on the effectiveness of the identified solution is left to reviewer expertise. Consider if sufficient effort is shown to properly preserve the software and allow its use in the far future (e.g. using services such as Software Heritage).

Rationale: If software is required, it must be preserved. Detailing how at the level of the DMP increases the likelihood that the issue will be addressed properly during the project.

Fulfilled Examples	Not fulfilled examples
• The software “readVoxels”, required to visualize and process the MRI voxel data will be containerized and shipped alongside the data proper.	• No explanation on the need to store software is given. • One or more required software are not detailed. • Software to access the data will be provided upon request (too generic).

[B.volume.1] Gives details of data volumes (bytes or numbers)

Scored as either 0 (not fulfilled), 0.5 (partially fulfilled) or 1 (fulfilled).

Explicitly details the volume of the generated data, even in broad strokes.

Rationale: Data volume informs a lot of storage and preservation choices. It may also warrant discussing specific storage solutions during active data management if big data is being processed.

Fulfilled Examples	Partially Fulfilled Examples	Not fulfilled examples
• MRI data will be expected to take between 3 to 5 terabytes of storage. • The project will create bibliographies of negligible size (1Mb).	• Details are not given for all data types. • Provenance will be tracked using specific methodology (too little information).	• No reference to data provenance.

[C.meta.1] Indicates which metadata will be provided

Scored as either 0 (not fulfilled), 0.5 (partially fulfilled) or 1 (fulfilled).

Explicitly lists which metadata will be provided for the data. This criterium is purposefully generic as to allow less mature DMPs to be scored for a preliminary work regarding metadata. Also fulfilled by stating that ontologies and vocabularies will be searched for to best fit the (meta)data that is being collected, especially if the way this search will be performed is illustrated. This is a “soft” version of C.meta.2.

Rationale: A list of required metadata for all data types is required, but it is sometimes hard to provide them in high detail while the DMP is being redacted.

Fulfilled Examples	Partially Fulfilled Examples	Not fulfilled examples
• The usage of community-accepted data vocabularies will be considered. The FAIR sharing database will be searched for suitable resources, or a new controlled vocabulary will be developed.	• Details are not given for all data types.	• No reference to metadata structure is provided.

[C.meta.2] Indicates metadata standards for each data kind or type

Scored as either 0 (not fulfilled), 0.5 (partially fulfilled) or 1 (fulfilled).

Explicitly lists all metadata for each (expressed) data type/format combination or, if these are not given, for each data type. Multiple data type/format combinations may use the same metadata schema. Also fulfilled if metadata schemas or existing ontologies are used. The relevance and sufficient specificity of the proposed metadata is left to reviewer expertise. This is the “hard” version of C.meta.1.

Rationale: It is important to decide which metadata to record for each data type before data is acquired. Some kinds of data cannot be saved after data is collected(or get increasingly harder to be recorded).

Fulfilled Examples	Partially Fulfilled Examples	Not fulfilled examples
• For MRI voxel data, the following metadata items will be recorded: operator name and OrcID, ...	• Details are not given for all data types.	• No reference to metadata structure is provided. • The usage of community-accepted data vocabularies will be considered. The FAIR sharing database will be searched for suitable resources, or a new controlled vocabulary will be developed.

[C.meta.3] Indicates which metadata standards will be used

Scored as either 0 (not fulfilled), 0.5 (partially fulfilled) or 1 (fulfilled).

Specifically indicates if existing (and potentially commonly used) ontologies or controlled vocabularies will be used. Indicates which existing vocabulary will be used, and how. Indicates which types from an ontology (or if whole ontologies) will be used. The relevance and sufficient quality of the proposed metadata standard(s) is left to reviewer expertise.

Rationale: Usage of standardized metadata schemas is essential for data reuse.

Fulfilled Examples	Partially Fulfilled Examples	Not fulfilled examples
For MRI voxel data, the International Standard Ontology for MRI metadata will be used in conjunction with the Dublin Core ontology. These metadata structures will be extended with 5 project-specific fields...	Details are not given for all data types.	No reference to metadata structure is provided.

[C.meta.4] Indicates metadata formats for all data types

Scored as 1 (fulfilled) or 0 (not fulfilled).

The format of the metadata is explicitly given. The specific format is not exactly important (see C.meta.5). If different metadata formats are given, which formats apply to which data type must be specified. If the metadata format is not determined for all kinds of data, then the criterium is not fulfilled.

Rationale: Metadata should be homogeneous throughout the project, and follow a similar metadata format. This is also why this is a binary question.

Fulfilled Examples	Not fulfilled examples
- All metadata will be recorded as Markdown files. - Metadata for MRI scans will be recorded via a specific form and saved as JSON data. All other metadata will be recorded as TXT files.	No metadata format is given. - Metadata for MRI scans will be recorded via a specific form and saved as JSON data (not all data types are included)

[C.meta.5] Indicates machine actionable metadata formats for all data types

Scored as 1 (fulfilled) or 0 (not fulfilled).

All metadata formats which are specified are machine actionable. This criterium cannot be fulfilled if some metadata formats are missing (i.e. C.meta.4 is 0). Machine actionability of the format is left to the reviewer's expertise, but, indicatively, RDF, XML, JSON, and other structured formats for dictionary data are good machine-actionable metadata formats.

Rationale: Machine-actionable metadata is essential for interoperability and findability, and in general to follow the FAIR principles.

Fulfilled Examples	Not fulfilled examples
Metadata for MRI scans will be recorded via a specific form and saved as JSON data. All other metadata will be recorded as XML files.	No metadata format is given. - Metadata for MRI scans will be recorded via a specific form and saved as JSON data (not all data types are included) - All metadata will be recorded as Markdown files (Format is not machine actionable).

[C.meta.6] Addresses the use of community metadata standards

Scored as either 0 (not fulfilled), 0.5 (partially fulfilled) or 1 (fulfilled).

All metadata standard specified conform to community-individuated best practices. The decision of which standards fulfill such a criterium is left to the expert reviewer.

Fulfilled Examples	Partially Fulfilled Examples	Not fulfilled examples
For MRI voxel data, the International Standard Ontology for MRI metadata will be used in conjunction with the Dublin Core ontology. These metadata structures will be extended with 5 project-specific fields...	Details are not given for all data types. - For MRI voxel data, the following metadata items will be recorded: ... (Metadata specific to the project and not necessarily interoperable)	No reference to metadata structure is provided.

[C.org.1] Defines file naming conventions

Scored as 1 (fulfilled) or 0 (not fulfilled).

Clearly specifies file naming conventions used for each data type or throughout the project. If the data is stored in different locations (see D.stor.1), specifies where the naming conventions will apply. Reason: Especially in multi-partner projects, file naming conventions keep all homogeneous files recognizable and make knowing the structure and format of each file easier.

Fulfilled Examples	Not fulfilled examples
- Throughout the HOR project, files will follow the following naming convention:... (Includes a pattern description for each naming convention) - The naming convention will be used for all files stored in the central server. Files stored in auxiliary, local storage locations can follow any naming convention, although they must be renamed when stored centrally.	No file naming convention is given. - Files will be named with the date and kind of experiment they are sourced from (Too generic).

[C.org.2] Addresses standard folder structures

Scored as 1 (fulfilled) or 0 (not fulfilled).

Standard folder structure is defined for each storage location. Each folder naming convention is also specified. This may include folder and sub-folder structure. May also specify permissions for access, deposit and modification for each (sub)folder.

Rationale: Standardized folder structure significantly limits the possibility of losing data due to poor organization, especially if it is documented. It also makes every collaborator aware of the storage location for their data.

Fulfilled Examples	Not fulfilled examples
- The central storage folder will be subdivided in the following subfolder structure: Experiments (for experimental data), Protocols, ... - The “experiments” folder will contain one folder per experimental type, and each, in turn, will contain one folder per experimental run...	No folder naming convention is given. - Folders will be structured according to best practices (too generic).

[C.org.3] Addresses the need for version control

Scored as 1 (fulfilled) or 0 (not fulfilled).

Addresses the need of controlling the versions of files, especially if these files will need to be modified by different people. This is especially important if code is being produced. If the need for a version control system is found, also defines a robust (possible) solution for the issue, such as a Version Control System software. May or may not reference every data type/format combination.

Rationale: File versioning can quickly get out of hand if not addressed promptly and with a specific solution.

Fulfilled Examples	Not fulfilled examples
- File versioning for analysis code will be handled by git. GitLab will be used to centralize and allow collaboration by different members. - No file versioning is required by the project (with a valid reason – consider that most projects either contain code or at paper at the very least)	File versioning is not considered - Different versions of the same file will be tracked by appending progressive numbers at the end of their file names (versioning strategy is not adequate).

[C.org.4] Indicates which identifiers will be used for sample labelling

Scored as either 0 (not fulfilled), 0.5 (partially fulfilled) or 1 (fulfilled).

Specifically indicates which labelling strategy will be used for physical and digital samples. This refers specifically to sample, aliquot or observation identifiers. These identifiers may or may not be persistent or globally unique (see C.org.7), but must be unique in the context of the project. Identifies a solution to assure local uniqueness.

Rationale: Identification of samples is essential for data and metadata cross-referencing. Identifiers must be at least locally unique to be useful, and a centralized method to check this uniqueness is essential, especially if sampling occurs simultaneously.

Fulfilled Examples	Partially Fulfilled Examples	Not fulfilled examples
Each sample of concrete will be labelled as “YYYYMMDD{collection_site_id}{prog}”, with collection side IDs and a progressive number. A sample labelling table will be kept updated as new samples are collected.	Details are not given for all data types. Each sample of concrete will be labelled as “YYYYMMDD{collection_site_id}{prog}”, with collection side IDs and a progressive number. (Does not include a uniqueness check strategy)	No reference to sample labelling is given.

[C.inter.1] Uses interoperable identifiers for metadata

Scored as either 0 (not fulfilled), 0.5 (partially fulfilled) or 1 (fulfilled).

Similar to C.inter.2. The identifiers used for metadata definitions (e.g. column names) are interoperable. For example, they use URLs or semantics which refer to shared, community definitions. Also fulfilled if the problem is taken into consideration and no standard definitions are applicable (See also C.meta.2). The interoperability of any specific identifier is left to reviewer expertise.

Rationale: Usage of interoperable definitions for data characteristics is essential for FAIR Interoperability.

Fulfilled Examples	Partially Fulfilled Examples	Not fulfilled examples
The following metadata will be considered: data acquisition date (following ISO standards), organism ID (lowercase, underscore-separated, from the NCBI Taxonomy database), location data will use the Wikidata location IDs, ...	Only some metadata identifiers are listed, and/or only some identifiers are interoperable.	Metadata is not specified, or specific identifiers are not listed. Metadata identifiers are not interoperable.

[C.inter.2] Uses interoperable definitions for data

Scored as either 0 (not fulfilled), 0.5 (partially fulfilled) or 1 (fulfilled).

Similar to C.inter.1. The identifiers used for data definitions (e.g. column names) are interoperable. For example, they use URLs or semantics which refer to shared, community definitions. Also fulfilled if the problem is taken into consideration and no standard definitions are applicable (See also C.meta.2). The interoperability of any specific identifier is left to reviewer expertise.

Rationale: Usage of interoperable definitions for data characteristics is essential for FAIR Interoperability.

Fulfilled Examples	Partially Fulfilled Examples	Not fulfilled examples
The following standard for data definitions will be used: data acquisition date (following ISO standards), gene identifiers will use ENSEMBL identifiers, mass measurements will be normalized to kilograms, ...	Only some data definitions are listed, and/or only some of them are interoperable.	Data fields are not specified, or specific definitions are not listed. Metadata identifiers are not interoperable.

[C.inter.3] Uses persistent, resolvable identifiers and an ID minting service for (meta)data

Scored as either 0 (not fulfilled) or 1 (fulfilled).

When new identifiers are minted, a explicitly defined ID minting service is used to generate them and make them resolvable, if possible. Which identifiers can be made persistent and resolvable is left to reviewer expertise. This criterium refers to data or metadata, not whole datasets: DOI minting of whole publications or datasets are not sufficient to fulfills this criterium.

Rationale: When possible, persistent, resolvable, globally unique identifiers (such as DOI) are to be preferred over local-only identifiers, as they increase the findability and interoperability of the data.

Fulfilled Examples	Not fulfilled examples
Each sample ID will be linked to a W3C ID resolver service which points to its metadata in JSON format.	Sample IDs are not considered The services specified are not sufficiently robust and persistent - Data will be deposited in Zenodo, giving a DOI to each dataset (Not specific enough to - Different versions of the same file will be tracked by appending progressive numbers at the end of their file names (versioning strategy is not adequate).

[C.inter.4] Avoids storing metadata exclusively in file names or labels

Scored as either 0 (not fulfilled) or 1 (fulfilled).

All metadata present in file names, labels or identifiers is also recorded as their own metadata fields.

Rationale: Relying on file names to recognize data and metadata is useful during the project but makes the (meta)data less machine actionable and more easily subjected to errors.

Fulfilled Examples	Not fulfilled examples
Samples will be labelled with date, sample type, experimental condition and operator id. (Statements can be far apart) Metadata regarding samples will contain: date, sample type, experimental condition, operator id, ...	(Meta)data semantics are not discussed. There is no redundancy between (meta)data content and its file name.

[C.meta.7] Considers extra metadata specifically required for reuse

Scored as either 0 (not fulfilled) or 1 (fulfilled).

Explicitly considers if additional metadata is required to promote data reusability in other contexts.

Rationale: Data and metadata useful for the current project may still be too restrictive to achieve effective sharing.

Fulfilled Examples	Not fulfilled examples
While not relevant to the project, data regarding environmental conditions around the beetle collection areas (air quality, wind speed, altitude) will be collected to promote data reuse. Beetle color will also be recorded.	No additional (meta)data collection is planned.

[C.quality.1] Indicates how (meta)data will be gathered

Scored as either 0 (not fulfilled), 0.5 (partially fulfilled) or 1 (fulfilled).

Explicitly lays out a strategy for (meta)data collection and recording (e.g. forms, reference manuals, guidelines, etc...) which will be known by all operators that will collect (meta)data. This includes all processes that take a trace, store it and preprocess it (up to “preprocessed data” in the data lifecycle). These protocols may be deferred to external documentation, or might be created during the project. Therefore, a statement indicating that they will be created is sufficient.

Rationale: Especially for large-scale projects, standardized (meta)data collection protocols are essential to ensure homogeneous, standardized and controlled (meta)data creation. They also serve as useful metadata for reuse.

Fulfilled Examples	Partially Fulfilled Examples	Not fulfilled examples
- A detailed biopsy collection manual with all collection procedures will be created and given to all sample collection pathologists. - Patient anamnesis forms will be used to collect patient-related metadata. They will contain the following information: ...	Only some data types are covered.	Information on (meta)data gathering protocols is not given. - Data will be collected using industry-standard best practices (too generic).

[C.quality.2] Indicates who is responsible for gathering which (meta)data

Scored as either 0 (not fulfilled), 0.5 (partially fulfilled) or 1 (fulfilled).

For each data type and collection protocol, specified who is going to perform that kinds of data collections or, alternatively, who is responsible for ensuring the quality of data and metadata collection during the performance of that protocol. The name, surname and contact information of those involved should be included.

Rationale: Having specific people to contact if and when problems or inconsistencies arise when data is processed or reused increases the likelihood of fixing them.

Fulfilled Examples	Partially Fulfilled Examples	Not fulfilled examples
J.M. (OrcID) will perform data collection protocols A through D. D.F. (OrcID) will perform	Only some data types are covered.	- Information on (meta)data gathering protocols is not given. - Data will be collected using industry-standard best practices (too generic).

[C.quality.3] Specifies which (meta)data gathering forms will be used

Scored as either 0 (not fulfilled), 0.5 (partially fulfilled) or 1 (fulfilled).

This is a more stringent version of [C.quality.1]. Other than simply describing the processes used to collect the data, specific (meta)data gathering forms are also (planned to be) used during data gathering processes. Some kinds of data cannot be collected through standard forms, or are collected automatically and therefore are not considered for the purposes of this criterium. Which kinds of data require form and which do not is left to reviewer expertise. These forms can be separately stored and updated independently of the DMP (but the details of the two must remain synchronized).

Rationale: Standard (meta)data gathering forms ensure that data and metadata are collected in an homogeneous way, ensuring their quality and, in general, helping upholding all other RDM good practices.

Fulfilled Examples	Partially Fulfilled Examples	Not fulfilled examples
A standard data gathering form will be used to collect patient metadata. This form, together with information on how to fill it, is available here (with a link, preferably a PID).	Only some data types come with predetermined data collection forms.	No (meta)data collection forms are given.

[C.quality.4] Indicates how data quality will be assessed

Scored as either 0 (not fulfilled) or 1 (fulfilled).

Considers whether data quality will be checked, and if so, how. This may be done through data ingestion protocols, specific people that will provide testing, or other automatic, semiautomatic or manual procedures. The same criterium for metadata is given in C.quality.5.

Rationale: It is important to consider how data is checked as it is collected and stored throughout the project, as it makes data deposition much faster and can catch errors or other problems (which may warrant the modification of the DMP) before they become unsurmountable.

Fulfilled Examples	Not fulfilled examples
• J. S. will develop ad-hoc data quality check pipelines to ensure that the collected data fulfills the requirements as set out in this DMP. • As the data is stored in a PostgreSQL database, data format and other checks will be performed automatically thanks to column and table rules.	• No quality assurance methodology is given. • Only some data types are quality checked.

[C.quality.2] Indicates how metadata will be checked and validated

Scored as either 0 (not fulfilled) or 1 (fulfilled).

Considers whether metadata quality will be checked, and if so, how. This may be done through metadata ingestion protocols, specific people that will provide testing, or other automatic, semiautomatic or manual procedures. The same criterium for data is given in C.quality.4.

Rationale: It is important to consider how metadata is checked as it is collected and stored throughout the project, as it makes data deposition much faster and can catch errors or other problems (which may warrant the modification of the DMP) before they become unsurmountable.

Fulfilled Examples	Not fulfilled examples
• J. S. will develop ad-hoc metadata quality check pipelines to ensure that the collected data fulfills the requirements as set out in this DMP. • As the metadata is stored in a PostgreSQL database, data format and other checks will be performed automatically thanks to column and table rules.	• No quality assurance methodology is given. • Only some metadata types are quality checked.

[C.quality.3] Indicates if calibration is needed and how it will be done

Only applicable if instruments are used in the project (e.g. not relevant for purely theoretical work).

Scored as either 0 (not fulfilled), 0.5 (partially fulfilled) or 1 (fulfilled).

Some instruments require calibration using standards, either internally or externally sourced. This criterium is fulfilled if either standards will be used and the DMP states how, or, alternatively, if instrument calibration is handled by some other party or is assured to be satisfactory thanks to some process external to the project. The distinction of which instruments require calibration and which do not is left to reviewer expertise. However, consider official guidelines such as ISO 9001 (Clause 7.1.5, **Monitoring and Measuring Resources**) to inform the decision.

Rationale: Instruments must be calibrated properly to ensure data reliability and quality.

Fulfilled Examples	Partially Fulfilled Examples	Not fulfilled examples
• Micropipettes are routinely checked by the laboratory staff and thus require no particular calibration for this project. • Calibration of the electrical frequency generators and counters will be performed by International Scientific Instruments Inc. before the project starts.	• Only some instruments that require calibration are taken in consideration.	• Instrument calibration is not taken into consideration.

[C.quality.7] IA - Indicates how measurement precision will be recorded

Only applicable if instruments are used in the project (e.g. not relevant for purely theoretical work).
Scored as either 0 (not fulfilled), 0.5 (partially fulfilled) or 1 (fulfilled).

The handling and recording of instrument (in)precision and errors is explicit. At minimum, recording of the instrumental error is done at the level of the instrument themselves.

Rationale: Instrumental error must be recorded in order to properly deal with it throughout data analysis. When sharing data, errors related to it are useful to inform reuse.

Fulfilled Examples	Partially Fulfilled Examples	Not fulfilled examples
• Spectrophotometers used have a detection limit to two decimal places. No measurement error is inherent to in-field species counting. • Micropipette sampling errors are ignored as is routine when considering cell cultures.	• Measurement error is taken in consideration only for some kinds of instruments.	• Measurement error is not taken into consideration.

[D.stor.1] Indicates where data will be stored during the project

Scored as either 0 (not fulfilled) or 1 (fulfilled).

Explicitly states which solution or solutions will be employed for data storage during the project. Explains why this specific solution or solutions have been selected. Moreover, these solutions fulfill minimum usability requirements posed by the handled data (in terms of volume, privacy sensibility, legal requirement, access permission needs, etc...). The judgement on suitability of a certain solution to the data handled by the project is left to reviewer expertise.

Rationale: The choice of data storage is essential for proper RDM and underlines all other choices.

Fulfilled Examples	Not fulfilled examples
Data storage facilities will be provided by the hosting university ICT department. They allow multi-user access and permission management, are routinely backed up and follow industry-standard security practices. They are suitable for storing the amount of data expected to be produced by the project.	No solution is identified. - Data will be stored in Zenodo (The determined solution is not suitable for handling hot data).

[D.stor.2] Addresses the need of a (meta)data backup policy during the project

Scored as either 0 (not fulfilled) or 1 (fulfilled).

The backup policy of the (meta)data is explicitly outlined. Backup copies of the (meta)data might or might not be suitable for all kinds of data, but, if they are not possible, other bit-level preservation strategies should still be delineated. These efforts can be offloaded to the provider of the data storage solution (see D.stor.1). This criterium is also fulfilled if no backups are required, although why this is the case should be properly argued.

Rationale: Data backups are essential to avoid data loss due to damages or accidental deletion or modification, especially for high-value data which is impossible or costly to replace.

Fulfilled Examples	Not fulfilled examples
Data storage facilities will be provided by the hosting university ICT department. They allow multi-user access and permission management, are routinely backed up and follow industry-standard security practices. They are suitable for storing the amount of data expected to be produced by the project.	No solution is identified. - Each collaborator is in charge of their own data backup policies (Solution is not satisfactory).

[D.stor.3] Addresses the need of redundant (meta)data storage locations

Only applicable if highly sensitive, extremely endangered data is handled. Scored as either 0 (not fulfilled) or 1 (fulfilled).

Explicitly addresses if the data is critically important and requires separate data storage locations for redundancy and infrastructural resilience. If this is the case, also defines how this will be achieved.

Rationale: Extremely important or sensitive data must also be resilient to datacenter failure, requiring multiple copies in different infrastructures.

Fulfilled Examples	Not fulfilled examples
Critical data regarding the deployment of armed forces in active conflicts will be stored in two separate locations: ... (with details of the two locations, and how to keep them synced).	Infrastructural redundancy is not taken in consideration.

[D.access.1] Indicates who will have access to the data during the project

Scored as either 0 (not fulfilled) or 1 (fulfilled).

Indicates roles of data handling individuals and which permissions they will have in accessing, creating, modifying and deleting the project's data. Indicates how these roles will be implemented in the data storage solution (see D.stor.1). If all individuals are given administrator-level permissions, details why it is safe and/or required to do so.

Rationale: Administrator-level access to all the data by all individuals is not advised as accidental deletions, modifications and other form of data mismanagement are more likely to occur. These problems are reduced with the creation of roles with minimal permissions for each different user.

Fulfilled Examples	Not fulfilled examples
- As the project involves only 3 individuals, all of them will have unrestricted access to the data. - Four different data-handling roles will be created in the platform: Administrator, Data Curator, Uploader and Reader. They will have the permissions as seen in Table ...	Roles and permissions are not identified. - Every researcher will have unfettered access to all data (No sufficient explanation of general admin access).

[D.access.2] IA - Indicates for how long data will be accessible to each partner

Only applicable to multi-partnership collaborations (i.e. with more than one institution). Scored as either 0 (not fulfilled) or 1 (fulfilled).

Defines for how long each partner institution will have access to which data. Also determines how access control will be achieved.

Rationale: Similar to D.access.1. Additionally relevant if institutions are in different legislative areas and therefore data transfer agreements must be signed.

Fulfilled Examples	Not fulfilled examples
The heading institution will have permanent access to all the data throughout the project. Partner institution A will access the data for the first two years of the project, until Deliverable 3 is achieved.	Data access by partner institutions is not delineated.

[D.safety.1] IA - Indicates which data protection laws apply to the data

Up to reviewer expertise, can be not applicable if it is extremely obvious that no data protection laws apply to the data handled by the project. Scored as either 0 (not fulfilled) or 1 (fulfilled).

Addresses explicitly if any data protection laws apply to the data, and if so, which.

Rationale: Explicitly referring to which data protection laws are involved in data handling makes everyone aware of their presence and gives them a chance to familiarize with them.

Fulfilled Examples	Not fulfilled examples
• No data protection laws apply to the data handled by the project. • The European General Data Protection Regulation (GDPR) applies to the data. • Data is not covered by any law, and it is very obvious that this is the case, up to reviewer discretion.	• It is unclear if data is covered by existing laws and regulations, and it is not explicitly discussed.

[D.safety.2] Addresses data protection, especially if data is sensitive

Scored as either 0 (not fulfilled) or 1 (fulfilled).

Addresses how data will be protected from unauthorized access, modification and deletion. Measures should be proportionate to the sensitivity of the data. If human data is handled, addresses specifically all different data storage locations (including laptops, USB stick and the like) for possible security issues and related mitigation solutions.

Fulfilled Examples	Not fulfilled examples
• Data is not particularly sensitive, so no data protections will be in place other than the requirement for users to log in to the shared data storage location. • Data will not be accessible over the internet, requiring users to log in to a local terminal for access. Log-ins will be tracked and require two-factor authentications. All database transactions will be logged for security purposes.	• Data protection is not addressed.

[D.safety.3] Describes the main risks related to data safety/dual use

Scored as either 0 (not fulfilled), 0.5 (partially fulfilled) or 1 (fulfilled).

For each data type, considers privacy and other kinds of risks as well as dual use implications.

Rationale: It is important to recognize data safety risks and the possibility of dual use before starting data handling. While this is not a formal DPIA, these considerations might lead to a formal risk assessment analysis.

Fulfilled Examples	Partially Fulfilled Examples	Not fulfilled examples
• The project will handle AIDS patient personal data. Unauthorized access to the data might impact individual liberties and future opportunities. • The project will handle migratory data of endangered bird species. As such, data might be exploited by poachers and will therefore be treated as potentially dangerous.	• Only some data types are addressed.	• Data risks are not taken into consideration.

[D.safety.4] Addresses the main risks related to data safety/dual use

Applicable only if risks have been found in D.safety.3. Scored as either 0 (not fulfilled), 0.5 (partially fulfilled) or 1 (fulfilled).

If data security risks have been found, determines methods that avoid or reduce them. If risks have not been taken into consideration, score this as 0.

Rationale: It is important to recognize data safety risks and the possibility of dual use before starting data handling. Providing safeguards against these risks and minimizing them is essential.

Fulfilled Examples	Partially Fulfilled Examples	Not fulfilled examples
• A full DPIA with data security experts will be performed and all data security issues will be addressed. • Data will be anonymized immediately after collection, reducing the risk of it being disseminated.	• Only some data types are addressed.	• Measurement error is not taken into consideration.

[D.safety.5]

Applicable only if risks have been found in D.safety.3. Scored as either 0 (not fulfilled) or 1 (fulfilled).

If data security risks have been found, outlines relevant data protection policies of all the institutions involved in the project, with relevant links. Briefly summarizes their content. If risks have not been taken into consideration, score this as 0.

Rationale: Institutional data protection policies define what can and cannot be done, as well as which services can or must be contacted in the various institutions before starting data handling.

Fulfilled Examples	Not fulfilled examples
• No institutional guidelines are present that address data security. • The university has one data protection guideline, defining two specific services to contact before handling sensitive data ...	• Data protection is not addressed. • Possible institutional policies are not addressed.

[E.privacy.1] Indicates if GDPR-sensitive data is handled

Scored as either 0 (not fulfilled) or 1 (fulfilled).

Indicates explicitly if privacy-sensitive human data (names, surnames, location information, contact information, pictures, medical data, genetic data, etc...) will be collected. For a definition of GDPR-sensitive data, refer to e.g. [this resource](#).

Rationale: GDPR-protected data must be handled in GDPR-compliant methodologies. It is important to understand if such data is being handled.

Fulfilled Examples	Not fulfilled examples
• No privacy-sensitive data will be handled by the project. • Some privacy sensitive data will be handled by the project: Names, surnames, dates of birth, social status.	• Data privacy issues are not addressed.

[E.privacy.2] IP - Indicates which legal basis is behind data handling

Applicable only if GDPR sensitive data is handled. Scored as either 0 (not fulfilled) or 1 (fulfilled).

Specifies under which legal basis data processing is being done. See GDPR Chapter 2, Article 6.

Rationale: A legal basis is required for lawful handling of GDPR-protected data.

Fulfilled Examples	Not fulfilled examples
• Personal data processing will be done only after obtaining informed consent and explicit permission from the subjects.	• No legal basis is defined.

[E.privacy.3] Indicates how the legal basis will be secured

Applicable only if GDPR sensitive data is handled. Scored as either 0 (not fulfilled) or 1 (fulfilled).

Specifies how the legal basis (see E.privacy.2) will be secured. See GDPR Chapter 2, Article 6.

Rationale: A legal basis is required for lawful handling of GDPR-protected data.

Fulfilled Examples	Not fulfilled examples
• Personal data processing will be done only after obtaining informed consent and explicit permission from the subjects.	• No legal basis is defined.

[E.privacy.4] Considers limiting handling of personal data during research

Applicable only if GDPR sensitive data is handled. Scored as either 0 (not fulfilled) or 1 (fulfilled).

The principle of minimization is one of the most fundamental principles of the GDPR. Explicitly states how the handled personal data must be handled for the purposes of the project, and/or defines when such data can be deleted.

Rationale: A legal basis is required for lawful handling of GDPR-protected data.

Fulfilled Examples	Not fulfilled examples
Personal data processing will be done only after obtaining informed consent and explicit permission from the subjects.	No legal basis is defined.

[E.privacy.5] Considers data obfuscation methods during research

Applicable only if GDPR sensitive data is handled. Scored as either 0 (not fulfilled) or 1 (fulfilled).

Sensitive data is protected even during handling via data obfuscation techniques (e.g. anonymization, pseudonymization). Not everyone in the project has access to personal data.

Rationale: Data obfuscation methods limit risks related to leaking personal data.

Fulfilled Examples	Not fulfilled examples
Personal data will be pseudonymized immediately after acquisition, and de-pseudonymization keys will be stored in a separate, secure location.	No obfuscation methods are considered.

[E.privacy.6] Identifies which data obfuscation methods will be used for preservation

Applicable only if GDPR sensitive data is handled. Scored as either 0 (not fulfilled) or 1 (fulfilled).

Explicitly addresses how GDPR-sensitive data will be obfuscated for preservation purposes. This often involved K-anonymity anonymization procedures.

Rationale: For preservation, data must be anonymized or treated similarly to avoid privacy risks.

Fulfilled Examples	Not fulfilled examples
Personal data will be pseudonymized immediately after acquisition, and de-pseudonymization keys will be stored in a separate, secure location.	No obfuscation methods are considered.

[E.privacy.7] IP - Indicates whether a DPIA is necessary or not

Applicable only if GDPR sensitive data is handled. To the discretion of the reviewer, this might also be scored as NA if a DPIA is not required, even though it is not explicitly listed. Scored as either 0 (not fulfilled) or 1 (fulfilled).

Explicitly addresses if a Data Protection Impact Assessment is required to properly handle personal data.

Rationale: DPIAs are sometimes required when dealing with highly sensitive personal data. It is important to recognize early if a DPIA is required, and who will carry it out.

Fulfilled Examples	Not fulfilled examples
• A DPIA will be carried out as we believe the risk involved in handling this personal data warrants its use. • As only basic information is collected (name, surname, contact email), a DPIA is not required.	• No DPIA is considered.

[E.privacy.8] IP - indicates who will carry out the DPIA

Applicable only if a DPIA is planned. To the discretion of the reviewer, this might also be scored as NA if a DPIA is not required, even though it is not explicitly listed. Scored as either 0 (not fulfilled) or 1 (fulfilled).

Identifies who will carry out the DPIA. If a DPIA is considered but it is ultimately not required, score this as fulfilled.

Rationale: DPIAs are sometimes required when dealing with highly sensitive personal data. It is important to recognize early if a DPIA is required, and who will carry it out.

Fulfilled Examples	Not fulfilled examples
• A DPIA will be carried out as we believe the risk involved in handling this personal data warrants its use. It will be carried out by the coordinating university legal and privacy office. • As only basic information is collected (name, surname, contact email), a DPIA is not required.	• No DPIA is considered. • Nobody is named if a DPIA is considered.

[E.owner.1] Indicates the owner(s) of the data

Scored as either 0 (not fulfilled) or 1 (fulfilled).

Indicates explicitly who is legally the owner of the data. This is especially important when human data is handled and, in any case, identifies who must be contacted if issues related to data arise. To be fulfilled, contact information of the owner must be provided. Note that the researcher(s) who generate the data might not be their owners, as contractual obligations might transfer their ownerships to e.g. their institutions. If data ownership is ambiguous, a data sharing agreement should be created to address it.

Rationale: A specific data owner is required to dispel any possible future disputes, legal or otherwise. It also makes it clear who must be contacted before publishing or sharing the data.

Fulfilled Examples	Not fulfilled examples
• The owner and controlled of all data produced by the project will be, jointly: H. B. Simmons (hbsimmons@institution.org), as principal investigator, and the University (contact@university.org).	• Data ownership is not taken into consideration.

[E.owner.2] Indicates access conditions applied to the data for preservation

Scored as either 0 (not fulfilled), 0.5 (partially fulfilled) or 1 (fulfilled).

Specifies if all data will be available for preservation, or if controlled or restricted access will be required. If not completely open, also specifies a reason for closing the data.

Rationale: It is important to specify if some data cannot be shared, and why it cannot be shared.

Fulfilled Examples	Partially Fulfilled Examples	Not fulfilled examples
• Data regarding MRI imagining can be openly shared. Anonymized patient clinical metadata will also be shared openly. • As no sensitive or protected data is handled by the project, all data will be shared openly. • Contractual obligations with BRB SpA does not allow data sharing at the end of the project. • Potential patent development will need to be assessed before allowing data dissemination.	• Only some data types are addressed. • Accessibility of data will be taken in consideration after the end of the project (Potentially correct, but requires handling the DMP later).	• No consideration on data accessibility is given.

[E.owner.3] Indicates which licenses will be applied to the data

Scored as either 0 (not fulfilled) or 1 (fulfilled).

Indicates which license(s) will be applied to the data when it is shared. Any license is suitable, but, as a general rule, CC0 is to be preferred to maximize reuse.

Rationale: Data licenses allow its reuse. Any data shared without a license, although open, cannot freely be reused.

Fulfilled Examples	Not fulfilled examples
• All shared data will be placed under the CC-BY license.	• Data licensing is not taken into consideration. • Not all open data types have a specified license.

[E.owner.4] Addresses the creation of consortium-level data ownership policies

Only applicable if a consortium (i.e. union of multiple institutions) is created to carry out the project. Scored as either 0 (not fulfilled) or 1 (fulfilled).

Indicates if a consortium-level formal data sharing agreement will have to be created, and if so, who is responsible to create it. This is most often done to delineate who is the owner of which dataset.

Rationale: Same rationale as E.owner.1, just applied to consortia.

Fulfilled Examples	Not fulfilled examples
A data sharing agreement will be created by the Principal Investigator to delineate how data will be shared and owned by the various consortium partners before month 6 of the project.	Data ownership is not taken in consideration. Nobody is given explicit responsibility in creating the agreement. Consortium-level data sharing policies are not delineated.

[E.owner.5] Addresses licensing issues (or lack thereof) of reused data

Scored as either 0 (not fulfilled) or 1 (fulfilled).

Considers if specific licenses are applied to the reused data, and how they will be followed during the project. This is especially important if Share-alike clauses are included in reused data. Score as fulfilled if one of the selection criteria for reusable data is that only CC-0 data will be used.

Rationale: Some licenses impose new limitations on data handling of new data, especially if they are combined with it.

Fulfilled Examples	Not fulfilled examples
- This project will consider for reuse only CC0 data. - No data can/will be reused, so there will be no issues regarding existing data licensing. - All data produced using CC-BY-SA data will also be made available openly under the same license.	Reused data licensing is not addressed.

[E.ethics.1] Addresses ethical concerns (or lack thereof)

Scored as either 0 (not fulfilled) or 1 (fulfilled).

Considers if the data may have ethical concerns regarding its acquisition or handling. Ethical concerns may arise due to the nature of the experiment, the nature of the subjects of the experiment or other circumstances.

Rationale: Knowing if ethical concerns might be involved in the running of experiments is a necessary prerequisite of handling them properly.

Fulfilled Examples	Not fulfilled examples
- As experiments involve macaques, ethical concerns are expected to arise. - As only cell lines will be handled during the project, no ethical concerns are expected.	Ethical concerns are not discussed.

[E.ethics.2] Addresses the need to contact the ethical board

Applicable only if ethical concerns are envisioned (see E.ethics.1). Scored as either 0 (not fulfilled) or 1 (fulfilled).

If ethical concerns might arise, considers if the ethical board must be contacted, and if so by whom, how, and with which deadline.

Rationale: Ethical boards must be contacted if ethics concerns have to be addressed.

Fulfilled Examples	Not fulfilled examples
The ethics board in the University will be contacted to give an opinion and approval of all experiments involving animals.	Ethical concerns are not discussed. Contacting ethical boards is not discussed.

[E.ethics.3] Addresses how ethic concerns affect data sharing

Applicable only if ethical concerns are envisioned (see E.ethics.1). Scored as either 0 (not fulfilled) or 1 (fulfilled).

If ethical concerns might arise, considers how these may affect data sharing and reuse. If possible, details methods to reduce or remove these limitations.

Rationale: Some ethical concerns (e.g. handling of protected species, specific wording of informed consent, de-anonymization concerns) might limit data sharing.

Fulfilled Examples	Not fulfilled examples
- As we believe that this data might potentially foster marginalization of a minority, we will not share it openly but only upon request. - The political survey data which will be collected will not be shared, even anonymously. This was done to prevent possible repercussions on the population.	Ethical concerns are not discussed.

[F.open.1] Describes which data can be shared and which can't

Scored as either 0 (not fulfilled), 0.5 (partially fulfilled) or 1 (fulfilled).

For each data type and format combination, details which data can be openly shared and which cannot be shared, or must be placed under and embargo or other access restrictions.

Rationale: Clearly stating which data will be shared promotes openness and requires follow-up at the end of the project.

Fulfilled Examples	Partially Fulfilled Examples	Not fulfilled examples
- All datasets will be openly shared after the end of the project. - All datasets will be shared openly, with the exception of the measuring apparatus blueprints, which will be placed under embargo for up to 4 years pending patenting.	Only some data types are addressed. - Sharing of data will be taken in consideration after the end of the project (Too vague).	No consideration on data availability is given.

[F.open.2] IA - Indicates why data cannot be shared openly

Only applicable if some (meta)data types cannot be shared. Scored as either 0 (not fulfilled), 0.5 (partially fulfilled) or 1 (fulfilled).

For each data type which will not be shared openly, specifies why this is the case. This does not include the metadata on the specific data type (see F.open.3).

Rationale: A reason to **not** share must be given to follow the “as open as possible, as closed as necessary” philosophy.

Fulfilled Examples	Partially Fulfilled Examples	Not fulfilled examples
All datasets will be shared openly, with the exception of the measuring apparatus blueprints, which will be placed under embargo for up to 4 years pending patenting.	Only some data types are addressed. - Sharing of data will be taken in consideration after the end of the project (Too vague).	No consideration on data availability is given.

[F.open.3]

Only applicable if some (meta)data types cannot be shared. Scored as either 0 (not fulfilled), 0.5 (partially fulfilled) or 1 (fulfilled).

For each data type which will not be shared openly, specifies why even metadata cannot be shared. Not fulfilled if even one metadata type which is not shared is not accompanied by an adequate description of the reasons for not sharing.

Rationale: A reason to **not** share must be given to follow the “as open as possible, as closed as necessary” philosophy. Metadata can (and should) usually be shared even if the data is under embargo or cannot be shared for some other reason.

Fulfilled Examples	Partially Fulfilled Examples	Not fulfilled examples
Some valid reason not to share metadata, which currently I cannot come up with.	Only some data types are addressed. - Sharing of data will be taken in consideration after the end of the project (Too vague).	No consideration on (meta)data availability is given.

[F.open.4] IA - Indicates how data sharing restrictions will be minimized

Only applicable if some (meta)data types cannot be shared. Scored as either 0 (not fulfilled) or 1 (fulfilled).

Explicitly indicates how data sharing limitations will be reduced or overcome. Also fulfilled if potential strategies to overcome these limitations will be tested after data collection. Reason: Methods like anonymization and summarization can be used to overcome most data sharing limitations. Requiring an explicit mention of possible ways to increase data sharing forces the authors to take it into consideration.

Fulfilled Examples	Not fulfilled examples
Data anonymization will be attempted at the end of the project to allow data sharing. However, at least k=10 will be required for security reasons. If this is not achievable, the data will be kept accessible only under request.	Data sharing limitations are not discussed. Ways to keep sharing data are not taken into consideration or planned.

[F.filter.1] Describes the process followed to decide which data to preserve

Scored as either 0 (not fulfilled) or 1 (fulfilled).

Specifies how data will be selected for preservation, or, alternatively, if all data should be preserved and why that is the case.

Rationale: Not all data must be preserved for long periods of time. As data preservation is a time-consuming, costly endeavor, selection of which data is worth preserving and for how long should be explicit. Note that replication of the study is almost always required, thus most generated data will need to be preserved, even for a limited period of time.

Fulfilled Examples	Not fulfilled examples
• Data regarding the detected economic trends is expected to be potentially useful in the future, even for historic reasons, so it will be preserved in perpetuity. Data regarding local stock exchanges will be preserved for 5 years for reproducibility purposes.	• Selection of data for preservation is not discussed. • All data will be preserved indefinitely (too vague).

[F.filter.2] Considers the target audience of reusers of the preserved data

Scored as either 0 (not fulfilled) or 1 (fulfilled).

Discusses who might be interested in reusing the shared data, other than for reproducibility purposes.

Rationale: Discussing this point explicitly forces the author to think why they are sharing the data, and for what (re-usability) purpose.

Fulfilled Examples	Not fulfilled examples
• Data regarding the detected economic trends could be reused by economists to generate new trend models, and by historians to describe the current economic growth rates.	• Selection of data for preservation is not discussed. • Description of potential target audiences for reuse is not given.

[F.filter.3] Indicates which data must be destroyed for legal reasons, and why

Only applicable if some (meta)data types cannot be shared. Scored as either 0 (not fulfilled), 0.5 (partially fulfilled) or 1 (fulfilled).

Discusses why some (meta)data must be destroyed for legal reasons and why this is the case.

Rationale: Some contractual or other legal obligations (e.g. too many privacy concerns) require data to be destroyed. Keeping this in mind and addressing it specifically can reduce the potential disruptive effect this destruction might have on using the data for the project and beyond.

Fulfilled Examples	Partially Fulfilled Examples	Not fulfilled examples
• Patient identification data will be destroyed as soon as pseudonymization is performed, as there is not need for it for the purposes of this project.	• Only some data types are addressed.	• No consideration on (meta)data availability is given.

[F.time.1] Indicates for how long data will be preserved

Scored as either 0 (not fulfilled) or 1 (fulfilled).

Defines for how long data will be preserved. For long-term data preservation, at least 13 years must be guaranteed. This is generally applicable to all data as a whole, so partial fulfillment does not apply. If data preservation length is specified for only some preserved data types, score this as not fulfilled.

Rationale: Not all data must be preserved for long periods of time. As data preservation is a time-consuming, costly endeavor, selection of which data is worth preserving and for how long should be explicit. Note that replication of the study is almost always required, thus most generated data will need to be preserved, even for a limited period of time.

Fulfilled Examples	Not fulfilled examples
• Data regarding the detected economic trends is expected to be potentially useful in the future, even for historic reasons, so it will be preserved in perpetuity. Data regarding local stock exchanges will be preserved for 5 years for reproducibility purposes.	• Selection of data for preservation is not discussed. • All data will be preserved indefinitely (too vague).

[F.time.2] Indicates when the data will be made available

Scored as either 0 (not fulfilled) or 1 (fulfilled).

Defines when (meta)data will be made available. This can happen as soon as the data is collected, at the end of the project, or after a (finite) predetermined duration. This is generally applicable to all data as a whole, so partial fulfillment does not apply. If data preservation length is specified for only some preserved data types, score this as not fulfilled.

Rationale: Determining in the DMP when the (meta)data will be made available makes the commitment tangible and the authors accountable for it.

Fulfilled Examples	Not fulfilled examples
• Data will be made available after publication (at most one year after the project ends). • Metadata regarding the collected data will be made available immediately after validation and quality assurance. Data, on the other hand, will be placed under a 5-year embargo.	• Dates of availability are not explicitly discussed.

[F.time.3] If delayed, indicates why the data will be embargoed

Only applicable if some data is not immediately made available. Scored as either 0 (not fulfilled) or 1 (fulfilled).

Specifies why (meta)data cannot be made immediately available.

Rationale: Considering this point explicitly should be done in order to follow the “as open as possible, as closed as necessary” philosophy.

Fulfilled Examples	Not fulfilled examples
• Metadata regarding the collected data will be made available immediately after validation and quality assurance. Data, on the other hand, will be placed under a 5-year embargo to allow the creation of scientific publications. After publication, embargo will end.	• Dates of availability are not explicitly discussed. • Metadata regarding the collected data will be made available immediately after validation and quality assurance. Data, on the other hand, will be placed under a 5-year embargo.

[F.open.5] If not via direct access, indicates how the data will be made accessible

Applicable only if direct download of (meta)data is not possible, but access is still given. Scored as either 0 (not fulfilled) or 1 (fulfilled).

Specifies which method(s) should be used to access data which is not available for direct download. If possible, also mentions the requirements needed by those asking for access to obtain it. If money is required for data transfer, also indicates why this is the case. This information can be omitted only if a completely automatic way to obtain (controlled) access is available, in which absolutely no human interaction is required). This is generally applicable to all data as a whole, so partial fulfillment does not apply. If access methods are specified for only some preserved data types, score this as not fulfilled.

Rationale: Accessibility is not necessarily only when data is directly obtainable. However, when this is not the case (need for data transfer agreements, sensitive data, large datasets), the specific way in which data can be obtained should be extremely clear.

Fulfilled Examples	Not fulfilled examples
• Data regarding migratory routes will be kept confidential for security concerns. It will be provided upon reasonable request for academic purposes. Details of reuse purposes must be given, and a data transfer agreement will be signed. • Data is too large to be uploaded to a repository for direct access. Access will be given by shipping copies of it in hard-drives or other physical support upon request.	• Accessibility methods are not discussed. • Data access will be given upon reasonable request (not enough information).

[F.open.6] If not shared openly, indicates who is responsible to keep the data Accessible for the long term.

Applicable only if direct download of (meta)data is not possible, but access is still given. Scored as either 0 (not fulfilled) or 1 (fulfilled).

Access methods for non-direct download data includes the names and contact information of those responsible to give access to the data in the long-term, or, alternatively, of the offices designated to carry it out. This information can be omitted only if a completely automatic way to obtain (controlled) access is available, in which absolutely no human interaction is required).

Fulfilled Examples	Not fulfilled examples
• Data regarding migratory routes will be kept confidential for security concerns. It will be provided upon reasonable request for academic purposes by contacting J.M Rodriguez (jm.rodriguez@institution.org) or the institutions RDM office (rdm@institution.org). Details of reuse purposes must be given, and a data transfer agreement will be signed. • Data is too large to be uploaded to a repository for direct access. Access will be given by shipping copies of it in hard-drives or other physical support upon request. Contact J.M Rodriguez (jm.rodriguez@institution.org) or the institutions RDM office (rdm@institution.org) for information and shipping and handling costs.	• Accessibility methods are not discussed. • Data access will be given upon reasonable request (not enough information). • Data regarding migratory routes will be kept confidential for security concerns. It will be provided upon reasonable request for academic purposes. Details of reuse purposes must be given, and a data transfer agreement will be signed (No contact information).

[F.findable.1] Indicates how the data will be findable by interested parties

Scored as either 0 (not fulfilled) or 1 (fulfilled).

Indicates how the (meta)data will be findable by others. This includes sharing the data in a searchable repository, using specific metadata, advertising the data on social media, data-specific publications in data journals, conference proceedings, etc... Sufficiency of this effort is left to reviewer expertise.

Rationale: Provisions to make potentially interested reusers aware of the availability of the data should be arranged to increase the value of the data and the probability of reuse.

Fulfilled Examples	Not fulfilled examples
• To increase findability, (meta)data will be indexed in ArchiRepo, the open repository of architectural information.	• Dates of availability are not explicitly discussed. • Metadata regarding the collected data will be made available immediately after validation and quality assurance. Data, on the other hand, will be placed under a 5-year embargo.

[F.findable.2] Indicates how PIDs will be used, and which PID service will be used

Scored as either 0 (not fulfilled) or 1 (fulfilled).

Indicates which Persistent Identifiers (PIDs) will be used, how, and which PID-minting service will be employed to generate them. If applicable, should also consider the cost of using such a PID service.

Rationale: Persistent Identifiers (PIDs) are a must when dealing with data, as the unambiguously refer to one and only one concept. Their thoughtful use is essential to both manage local data effectively and share it afterwards.

Fulfilled Examples	Not fulfilled examples
• The project will use locally minted PIDs for sample labelling, their minting provided manually via a master document. • The project will use DOIs for final deliverable, minted by doi.org.	• Usage of PIDs is not envisioned. • Permanent Identifiers will be used when possible (Too generic).

[F.findable.3] Indicates how data and metadata will be linked

Scored as either 0 (not fulfilled) or 1 (fulfilled).

Specifies the way in which metadata will be linked to the data and, if possible, a way to link the metadata to the data.

Rationale: FAIR principle F.3 “**Metadata clearly and explicitly include the identifier of the data they describe**” requires this step, but, in general, links from the metadata to the data should be present at the very minimum to increase data findability. Additionally, it makes it very difficult to mistake which metadata refers to which data, and vice-versa.

Fulfilled Examples	Not fulfilled examples
• Metadata will contain the resolvable PIDs of the data they refer to. Tabular data cannot contain backlinks, but they will be provided as deposition metadata.	• No discussion of data-to-metadata linking.

[F.findable.4] Indicates how metadata will be indexed and made searchable

Scored as either 0 (not fulfilled) or 1 (fulfilled).

Indicates an (online) service which indexes and allows searching through the metadata. This service should be available for the foreseeable future and provide a robust, complete search feature (i.e. other than title-based). Also details how the metadata will be supported by the platform hosting it, and to which degree it will be interoperable with other similar datasets.

Rationale: Metadata should be indexed and searchable, even if the data is not.

Fulfilled Examples	Not fulfilled examples
• Our metadata is mostly compatible with the BWX repository and will be indexed properly. However, as some fields are not supported, the whole metadata files will be reuploaded in their entirety to allow more complete indexing. • Zenodo allows a number of custom fields, however, the metadata interoperability of the platform is limited. We will collect all metadata into a single file to allow easier (local) searching.	• Metadata is not collected. • Discussion on the integration of the platform with the metadata is not done.

[F.findable.5] Identifies an accredited repository for data preservation

Scored as either 0 (not fulfilled) or 1 (fulfilled).

Identifies a satisfactory, accredited repository to preserve data. Arguments why this choice was made and outlines the related costs. Criteria for “satisfactory” and “accredited” are left to reviewer expertise, but consider relevance to the topic, scientific field, prestige and trustworthiness of the institution behind the repository, and more.

Rationale: A good data repository can make or break preservation efforts. It is important the sufficient thought and care is given when selecting one.

Fulfilled Examples	Not fulfilled examples
• As no relevant topical repository was found, we will use Zenodo as a data preservation service. As the volume of data is small, no costs are projected. • All sequencing data will be stored in the European Nucleotide Archive. All metadata required by the portal will be uploaded directly as such, and extra metadata fields will be deposited in Zenodo, containing backlinks to the ENA depositions.	• No discussion of repositories is done.

[G.resp.1] Addresses who is responsible for capturing data

Scored as either 0 (not fulfilled) or 1 (fulfilled).

Names the individual(s) which will perform and be responsible for acquiring and saving data.

[G.resp.2] Addresses who is responsible for assessing data quality

Scored as either 0 (not fulfilled) or 1 (fulfilled).

Names the individual(s) which will perform and be responsible for developing data quality guidelines (if not already done in the DMP), develop data quality assessment software and, in general, be responsible for data quality.

[G.resp.3] Addresses who is responsible for sharing data

Scored as either 0 (not fulfilled) or 1 (fulfilled).

Names the individual(s) which will perform and be responsible for sharing the data following the provisions of the DMP. These individuals are also those that can be contacted if the data was not shared in a timely or satisfying manner.

[G.resp.4] Addresses who is responsible for data safety and security

Scored as either 0 (not fulfilled) or 1 (fulfilled).

Names the individual(s) which will perform and be responsible for ensuring data safety and security.

[G.resp.5] Addresses who is responsible for data backup and integrity

Scored as either 0 (not fulfilled) or 1 (fulfilled).

Names the individual(s) which will perform and be responsible for performing data backups and ensuring integrity. If these services are offloaded to someone external, names the specific companies or institutions.

[G.resp.6] Indicates who is overseeing the implementation of the DMP

Scored as either 0 (not fulfilled) or 1 (fulfilled).

Names the individual(s) which will perform and be responsible for implementing the strategies defined in the DMP. Most often, this is the individual responsible for the project, the Principal Investigator.

[G.resp.7] Indicates who is responsible for reviewing and revising the DMP

Scored as either 0 (not fulfilled) or 1 (fulfilled).

Names the individual(s) which will perform and be responsible for updating the DMP, if it is needed. Most often this coincides with G.resp.6.

[G.resource.1] Indicates who will pledge time to data curation

Scored as either 0 (not fulfilled) or 1 (fulfilled).

Names the individual(s) which will perform and be responsible for curating data, including preprocessing, obfuscation, and other data-related functions outside of analysis.

Rationale: Data curation takes time and effort, especially if format modifications or controlled preprocessing has to be performed during the project or to enable data sharing.

[G.resource.2] Indicates which costs are related to data storage

Scored as either 0 (not fulfilled) or 1 (fulfilled).

Indicates the (projected) costs of hot data storage (i.e. storage during the project).

[G.resource.3] Indicates which costs are related to data preservation and sharing

Scored as either 0 (not fulfilled) or 1 (fulfilled).

Indicates the (projected) costs of cold data storage (i.e. storage after the project).

[G.resource.4] Indicates which costs are related to hiring data professionals

Scored as either 0 (not fulfilled) or 1 (fulfilled).

Indicates if hiring data curation professionals (i.e. data stewards) is required, and if so for how long and with which economic expenditure.